Predicting Breast Cancer Mortality Using Clinical and Genetic Profiles

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Repo link: https://github.com/mattsalgado721-bit/STAT_107_FINAL_PROJECT

ABSTRACT

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Breast cancer is a very complex disease with multiple variables that contribute to the recurrence of such cancer as well as mortality from the disease. In this project, our primary goal is to determine whether the likelihood of breast cancer recurrence and mortality can be predicted using a patient's genetic profile and clinical data. The current visual analysis using scatter plots, histograms, and box plots suggest that it is feasible to predict breast recurrence and mortality with the genetic and clinical data.

2. Introduction

For our project, we want to focus on the the question: using a patient's genetic and clinical profile, can we predict the likelihood of breast cancer recurrence or mortality? We'll be centering our research around specific genetic factors such as tumor mutational burden and clinical data such as tumor size, estrogen receptor status, and age at diagnosis. Our goal is to extract the data, perform various statistical procedures such as logistic regression to classify whether a patient is likely to experience a recurrence of breast cancer or death, similar to an actual medical research center.

3. Executive Summary

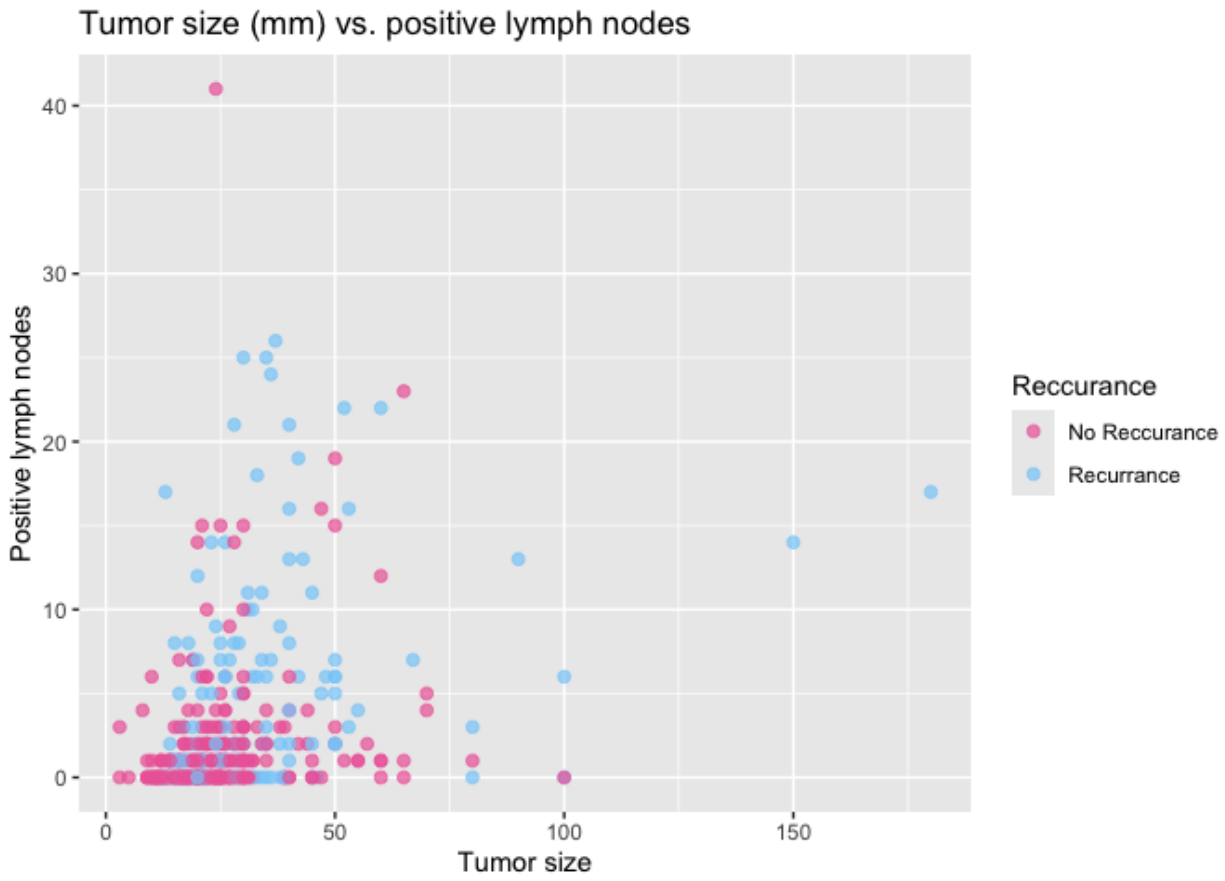
- We have found through scatterplots that when examining the count of patients' lymph nodes, there exists a positive and strong correlation with tumor size, suggesting that when there are higher counts of positive lymph nodes, the cancer itself is harder to contain, which may explain recurrences.
- Through histograms, we were able to identify a specific age range at diagnosis (years) that has the highest count of patients to further determine and narrow down when the recurrences are more likely to appear
- Alongside another histogram that helps us visualize the ranges of tumor sizes (mm) that the most patients logged, going hand in hand with the previous observation that the higher counts of positive lymph nodes and bigger tumor sizes are directly related, indicating a harder-to-contain cancer
- From the boxplots, we observed that mortality for breast cancer patients depends on factors such as tumor size (the smaller the tumor, the higher the chance of survival) and age (those below 70 years have a higher chance of survival), since breast cancer recurrence would only happen in those who have had it and eliminated it at least once before.

4. Data

- We sourced the data from cbioportal.org's targeted sequencing of primary breast tumors. After cleaning it up, our data set is 449 observations of 9 different variables.
- After cleaning up the data set, we decided on 9 variables to help answer our question. The variables and their significance are:
  - **ER_STATUS**: Whether or not the tumor cells have estrogen. These types of tumors respond best to hormonal therapy and generally have a better outlook for treatment
  - **GRADE**: The abnormality of the tumor under a microscope, a higher grade indicates faster growth and faster spread
  - **TMB_NONSYNONYMOUS**: Stands for tumor mutation burden and is a count of how many protein-changing mutations the tumor has
  - **TUMOR_SIZE**: The size of the tumor at the time of diagnosis in mm
  - **AGE_AT_DIAGNOSIS**: The age of the patient at the time of diagnosis
  - **LYMPH_NODES_EXAMINED_POSITIVE**: The number of nearby lymph nodes that contained cancer, a higher number indicates that the cancer has spread
  - **OS_BOOL**: Stands for overall survival status and is for if the patient has died. 0 means that the patient is living and 1 means that the patient has passed away
  - **RFS_BOOL**: Stands for recurrence-free survival status and is for if the cancer came back. 0 means that there was no recurrence and 1 means that there was a recurrence
  - **Recurrence**: Whether or not there was a recurrence of the cancer

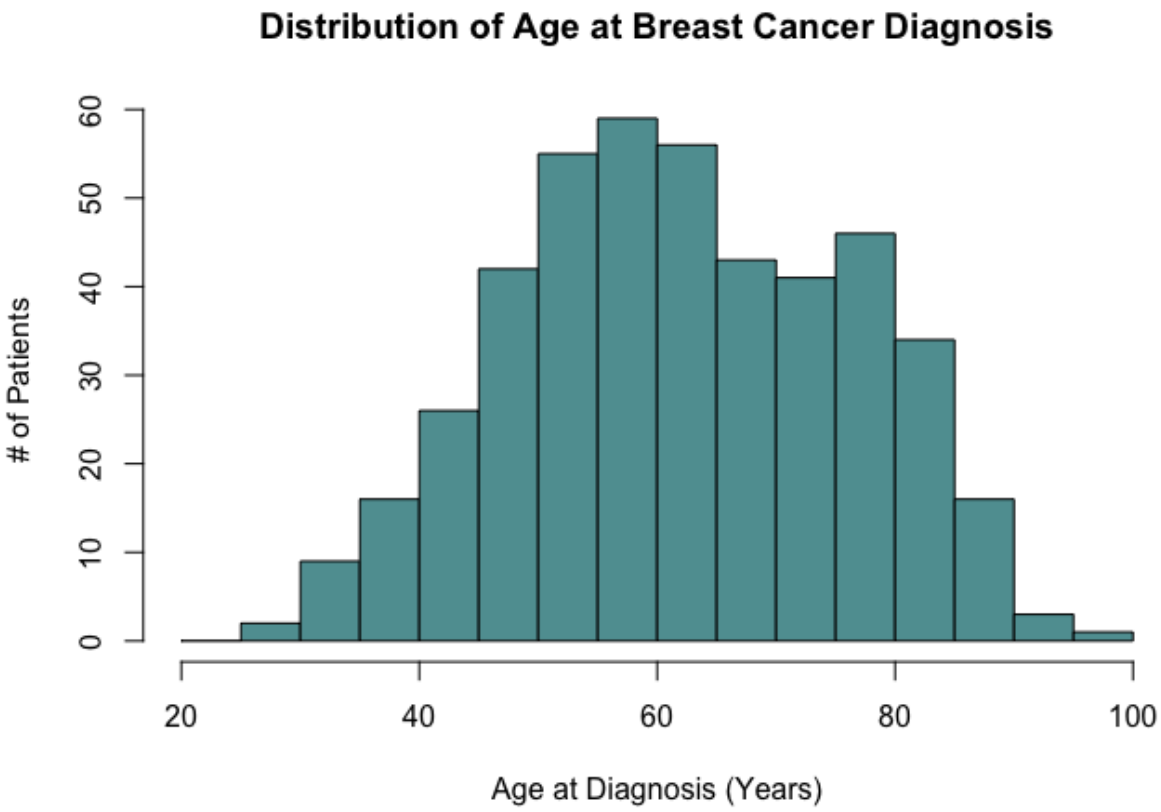
5. Visualization:

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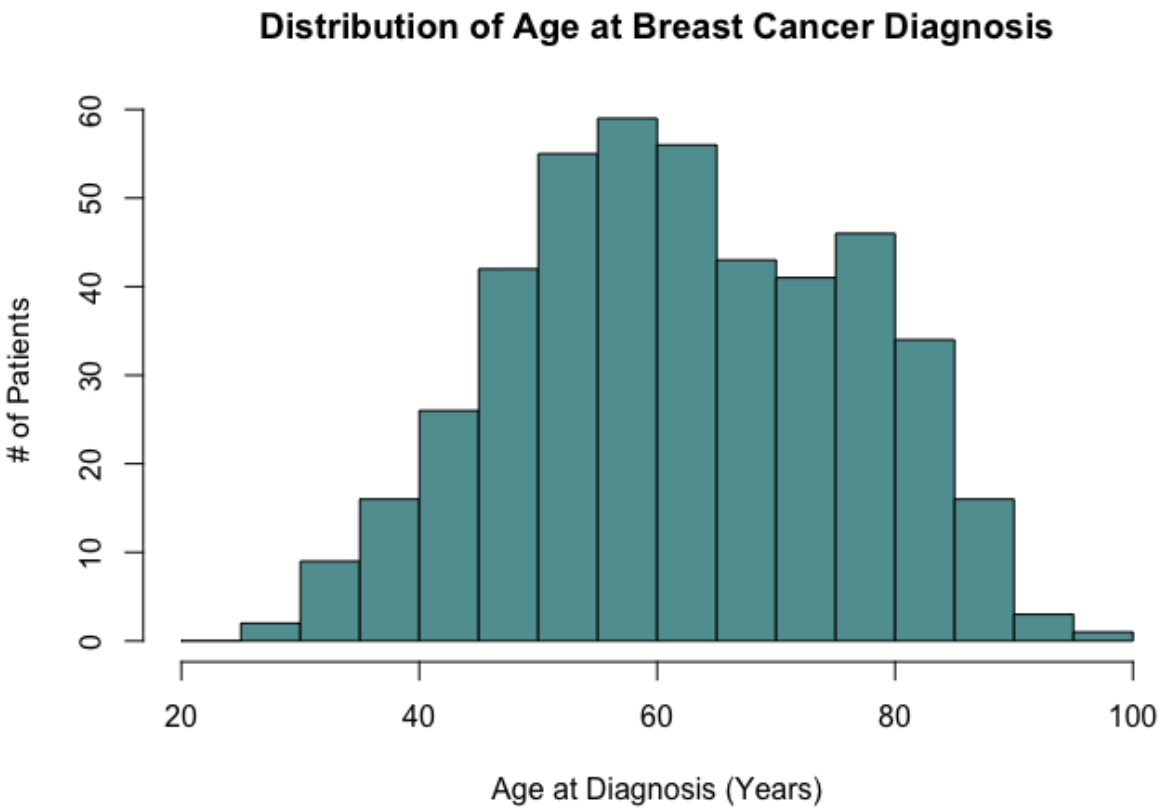
Our scatter plot of 'Tumor size vs. positive lymph nodes' with the data points differentiated by if there was recurrence or not suggest that recurrence is typically indicated by a larger tumor size and more positive lymph nodes.

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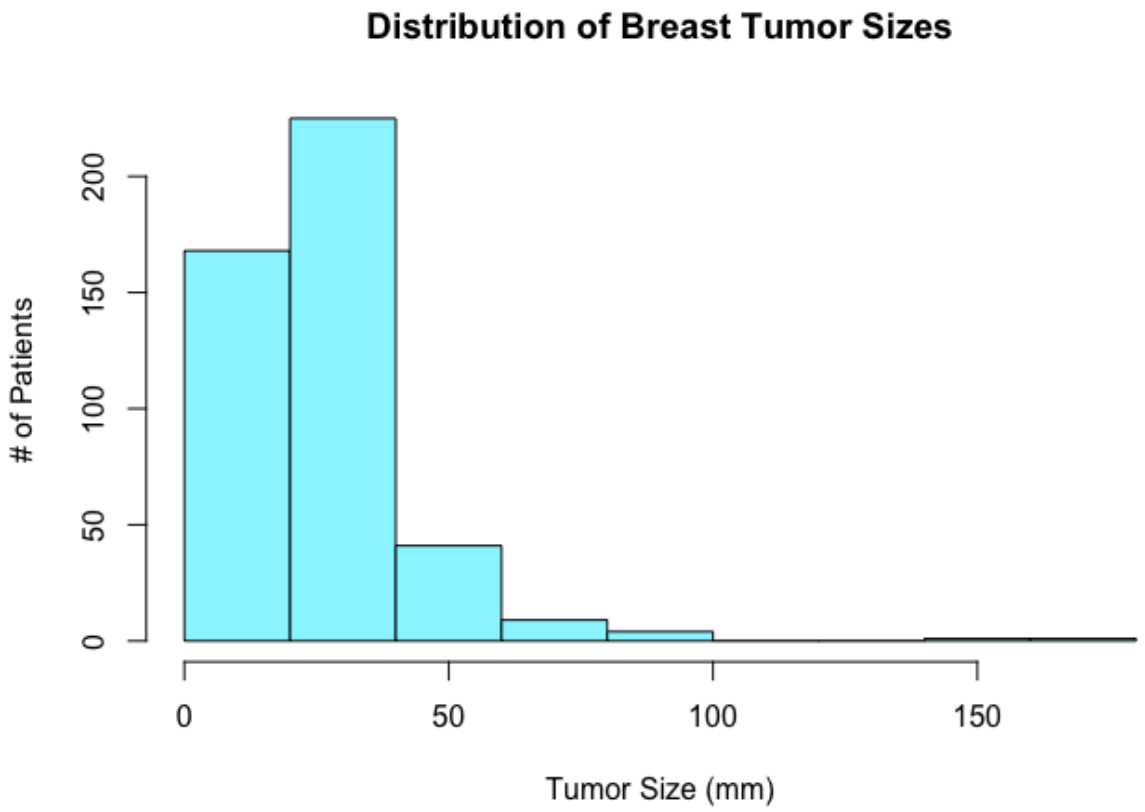
Moreover, the box plot 'Positive lymph nodes by recurrence' suggest that 'No recurrence' has a smaller cluster of positive lymph nodes compared to 'Recurrence' that has a much larger cluster of positive lymph nodes. This box plot suggests that the larger the number of positive lymph nodes, the more likely there is to be a recurrence of the cancer.

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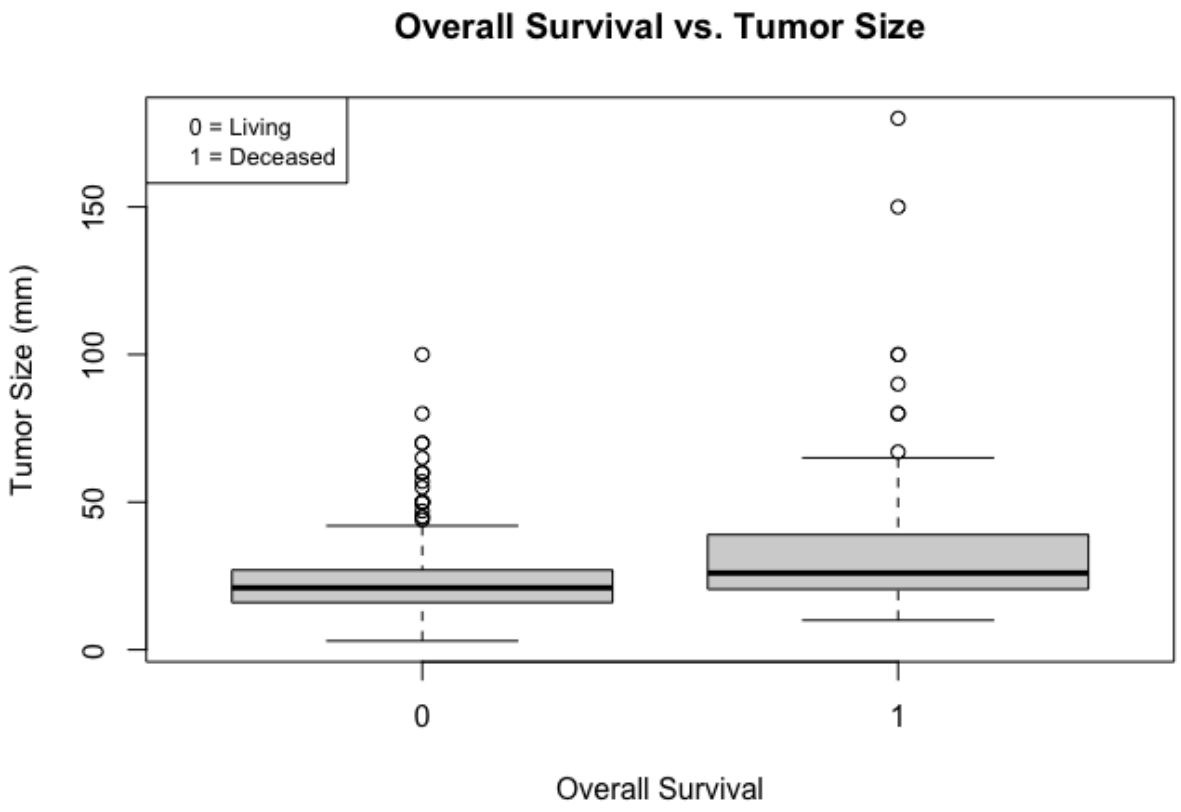
The first histogram represents the distribution of age at breast cancer diagnosis, with a range from 20-100 years and a spacing of 20 years. The shape seems to be fairly symmetrical and balanced and centered in the middle. Most patients were diagnosed between 45 and 85 years old. There don't seem to be any apparent outliers, the mode of the histogram for the relationship is uni-modal around 55-60 years with the number of patients reaching 60 within the sample.

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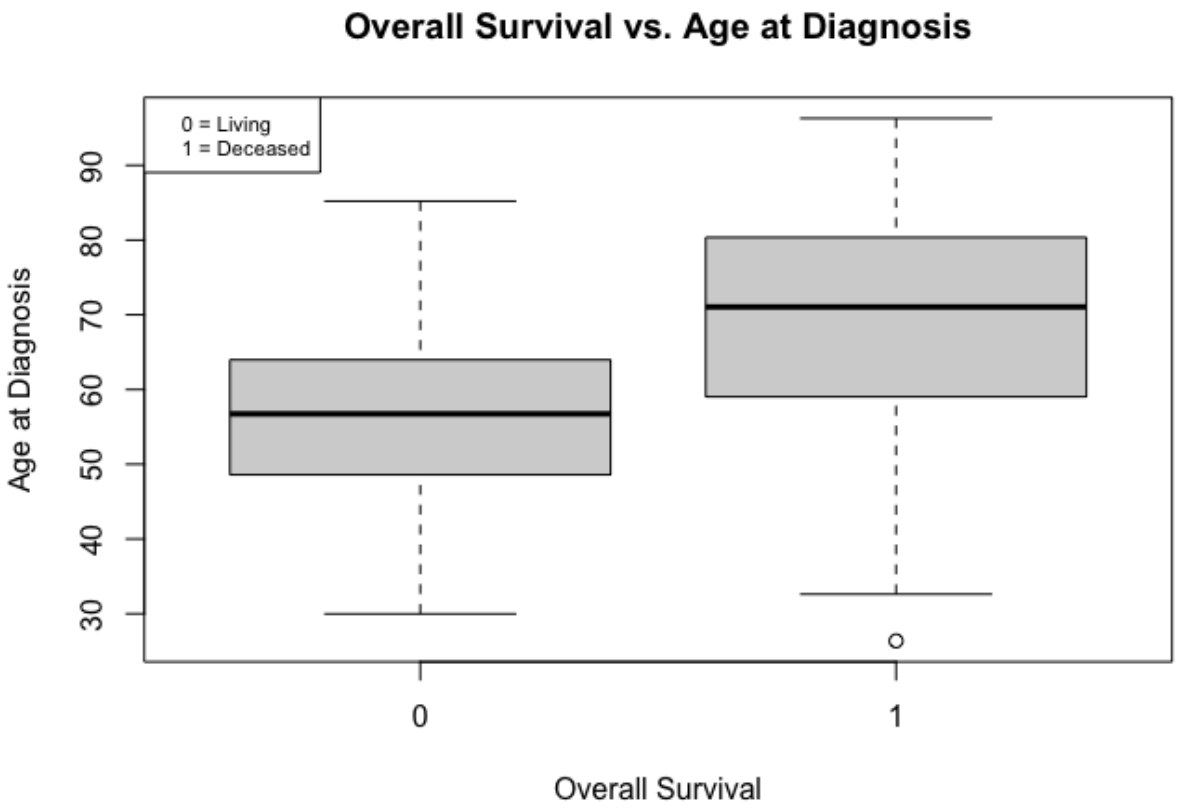
The second histogram represents the Distribution of breast tumor sizes diagnosis with a range from 0-150 mm with spacing of 50 mm, though most tumors seem to be below 50 mm. The highest number of patients have tumor sizes of 20-40 mm. The shape is strongly right-skewed and there don't seem to be many outliers; there is a possibility of larger tumor sizes around 150+ mm at the end of the histogram. The mode of the histogram for the relationship is 20-30 mm (unimodal) with the number of patients exceeding a count of 200 within the sample. Based on the histograms it is reasonable to assume that the chances of breast cancer reoccurring are higher if the individual is between ages 45-80, and tumor sizes between 20-30 mm are more apparent.

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From the box plot, the median tumor size for deceased patients is larger than the median tumor size for living patients. THE IQR for the deceased group, starts at a higher tumor size and is much larger than the IQR for the living group. We also observe that the biggest outliers for tumor size lay in the deceased group. This suggests that it would be feasible to use tumor size to predict whether a patient is likely to experience death.

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From the box plot, the median age for the deceased group is around 70 years while the median age for the living group is a little below 60 years. The IQR for the deceased group is at a higher position in the graph compared to the IQR for the living group. This suggest that age of diagnosis is feasible for use in predicting whether a patient is likely to experience death.

6. Analysis

We plan to fit logistic regression models with our data and examine which variables are the best predictors. We will do this by utiziling glm since our outcomes, mortality and recurrence, are binary variables.

7. Future work

- We need to further analyze the results and convert our predicted probabilities into binary classifications.
- This will let us see how well our model performs, and we can respond accordingly to the results.

Contributions

- Member 1 - Sri
  - Worked on the writing that explained our project details and created visuals ('Tumor size vs. positive lymph nodes' and 'Positive lymph nodes by recurrence') that could help describe the relationship between several variables. The scatter plots could serve as supporting visuals for why there may or may not be relationships between the variables and will be further analyzed with

models to better understand what the scatter plots could indicate. Defined the variables being used in the project and their significance to understanding our topic.

- Member 2 - Matthew Salgado
  - Worked on the data cleaning, the abstract, introduction, analysis, future work, and adding the visual images to the final report. Created the two box plots comparing tumor size with overall survival and comparing age at diagnosis with overall survival respectively.
- Member 3 - Janhavi Shelar
  - Worked on the executive summary for our observations and created histograms to model tumor sizes and the age demographic for patients diagnosed with summaries, and versions of the readme.